Portable, Low-cost, Gynecological Table to Improve Comfort & Increase Patient Retention of Cervical Cancer Exams in Cajamarca, Peru

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Introduction

Cervical cancer ranks as the ***second most common cancer*** among women in Peru. Peru also faces the ***sixth-highest mortality*** *rate* for cervical cancer globally [1].

Limitations of the current examination set-up used in gynecology clinics in Cajamarca, Peru (Fig. 1):

1. **Fixed, non-adjustable leg supports and lack of foot supports** [2]
   - Significant patient discomfort
   - Low retention rates for regular screenings
2. **Non-removable bottom flap** [2]
   - Healthcare providers have limited accessibility to examination areas
3. **Lack of Portability** [2]
   - Patients living in mountainous, rural regions have limited access to clinics



Figure 1. Existing gynecological examination table used in Cajamarca clinics with limited adjustability for patients of varying stature, and difficulties for providers to conduct examinations properly

Design Objectives

1. **Increase Patient Comfort**
   - Create leg supports with adjustable angles to reduce knee pressure
   - Design adjustable foot supports to accommodate various patient heights
2. **Improve Ease of Use for Provider**
   - Engineer mechanism to make bottom flap removable to improve provider access to examination areas
3. **Enhance Portability**
   - Make foot and leg supports detachable for out-of-clinic visits
4. **Ensure Sustainability**
   - Use locally available materials
   - Prioritize cost-effectiveness
   - Design for easy maintenance and repair

Results

Prototype Features and Functions

1. **Quarter Circle Angle Adjuster**
   - Includes continuous slot design for fluid adjustments and heat-set threaded insert for mounting existing leg supports obtained from Peruvian clinic, ensuring compatibility with local equipment
2. **U-shaped Swivel Mount**
   - Features continuous slot for smooth angle adjustments with integrated locking mechanism to securely fix leg supports in place, reinforced by thickened prongs and strategic fillets for optimal stress distribution
3. **Adjustable Sliding 80/20 Rail**
   - Utilizes dual 80-20 aluminum rails with 90-degree sliding brackets allowing both vertical and horizontal adjustments to accommodate various patient heights
4. **3D-printed Foot Support**
   - Incorporates rounded edges and soft foam padding to reduce pressure points and enhance patient comfort during examinations
5. **Height-adjustable Screws**
   - Provides multiple hole positions for flexible height adjustment while ensuring stable mounting and easy maintenance
6. **3D-printed PVC Anchor**
   - Maintains structural stability of leg supports through dual-point mounting and bottom anchor, while using readily available, low-cost PVC materials with simple 3D-printed components that can be easily reproduced locally

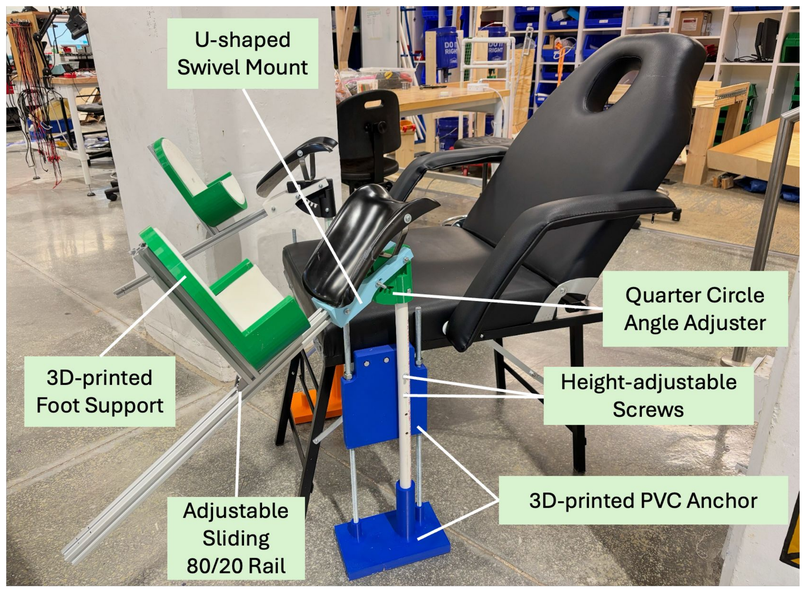


Figure 3. Final prototype design with leg stirrups and foot supports attached to chair

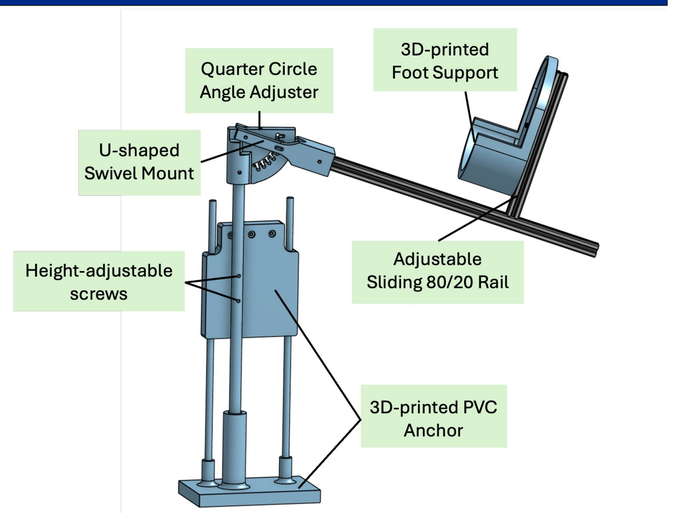


Figure 2. Computer Aided Design (CAD) of our integrated leg stirrup and foot support attached to anchor system for patient-specific adjustability

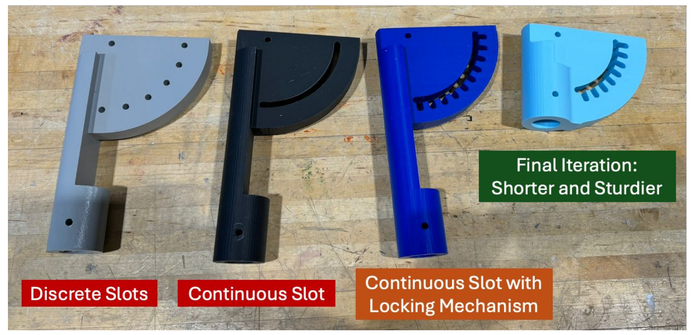


Figure 4. Iterations of quarter circle angle adjuster (left to right: oldest to newest)

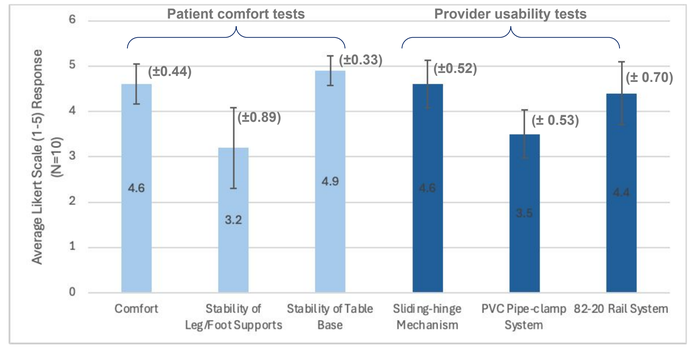


Figure 5. Likert-scale survey responses, error bars represent standard deviation across 10 trials (± SD)

Design Criteria and Testing

Design Criteria	Testing Procedure	Testing Results
Accommodates **average height** of Peruvian women [5ft 1in]	Evaluated stability of leg stirrups extended at min/middle/max positions with graduated weights	**Failed** — Successfully accommodates target height ± 5in, but prototype broke with heights beyond this range; also scored a low 3.2/5 on Likert-scale survey (Fig. 5);
Supports the weight of patients across a **range of body types** (BMI~26, safety factor = 2)	Test table strength with loads from 110-150 lbs with even weight distribution	**Passed** — Supported up to 150 lbs; 4.9/5 on Likert-scale survey (Fig. 5)
Minimizes knee pressure while maintaining **proper positioning** for gynecological exams	Evaluated comfort ratings from ≥ 10 volunteers testing for 5-minute periods	**Passed** — 4.6/5 on Likert-scale survey (Fig. 5)
Allows the provider to **adjust the angle** and **height** of the leg and knee supports **easily**	Evaluated setup time by multiple individuals, including people with and without a background in engineering	**Passed** —Positive feedback on ease of adjustments for variable patient heights (Fig. 5)
Uses materials **readily available** in Cajamarca or e**asily sourced locally** and within a budget of **$250**	Evaluated availability of materials from Peruvian suppliers, assessed cost of materials and procurement logistics	**Passed** — Total cost = $235.41
Uses materials **resistant to degradation** from medical disinfectants	Regular cleaning cycle and aggressive cycle with medical-grade disinfectants	**[Testing to be completed]**

Table 1: Design criteria brainstormed for proposed solution with appropriate testing plans carried out for each criteria

Conclusion

- Successfully designed an **easily adjustable** and **portable** examination table system with leg stirrups for gynecological exams in Cajamarca, Peru (Fig. 3)
- Achieved **high patient comfort** through adjustable supports and customizable positioning: 4.6/5 comfort rating on Likert Scale (Fig. 5)
- Demonstrated **strong technical performance**: 4.9/5 stability rating on Likert Scale (Fig. 5) and 300-lb weight capacity
- Identified **need for leg support stability improvement**: 3.2/5 leg stability rating on Likert Scale (Fig. 5)
- **Future steps:** Incorporate adjustable stirrups into a more size-appropriate table, implement provider feedback and prepare components for shipment to Cajamarca

References/Acknowledgments

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[1] J. S. Torres-Roman et al., “Cervical cancer mortality in Peru: regional trend analysis from 2008–2017,” Jan. 2021

[2] L. Vasudevan et al., “Barriers to the uptake of cervical cancer services and attitudes towards adopting new interventions in Peru,” Oct. 2020